

Week 05: Clear prompts and output checks

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Specify a task and validate an answer before using it.

Context and teaching plan

An office extracts amounts and invoice identifiers from synthetic invoices.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Task instruction

State exactly which fields to extract and how to handle missing entries.

Teacher question: Where does task instruction appear in the weekly case, and what could go wrong if we omit it?

Evidence

Use only the supplied invoice text; do not infer a value that is absent.

Teacher question: Where does evidence appear in the weekly case, and what could go wrong if we omit it?

Schema

Validate names and types of returned fields with Python.

Teacher question: Where does schema appear in the weekly case, and what could go wrong if we omit it?

Abstention

A missing value should stay null or cause a review flag, rather than be guessed.

Teacher question: Where does abstention appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Invoice INV-04 shows quantity 3 and price 12.
- Change or question: Extract ID and listed total if present.
- Supported observation: If total is absent, return None; 3×12 is a calculated check, not an extracted listed value.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The missing invoice total: An invoice has quantity 3 and price €12, but no printed total. Should the assistant fill €36 into the extracted total field?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Task instruction: analogy, mechanism and check

Familiar picture: A precise work order names each item and what to do if it is absent.

Familiar part	Technical meaning
Work order	prompt
Required items	fields
Missing-item rule	null instruction

Where this analogy breaks: A prompt guides behaviour but does not enforce correctness by itself.

Actual idea: State exactly which fields to extract and how to handle missing entries.

Work it aloud: Request `invoice_id` and `listed_total`; say “use null if the total is not printed.”

Ask before revealing: What should the output contain for this invoice’s listed total?

Supported response: null, because no printed total appears.

Look for this misconception: The assistant should infer every blank value.

Evidence: analogy, mechanism and check

Familiar picture: A clerk points to the exact line on a receipt before copying it.

Familiar part	Technical meaning
Receipt line	source evidence
Copied value	extracted field

Pointer	source span
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Where this analogy breaks: A source pointer may still point to the wrong line; verify the match.

Actual idea: Use only the supplied invoice text; do not infer a value that is absent.

Work it aloud: Quantity 3 and unit price €12 are visible. €36 is a calculation, not a printed total.

Ask before revealing: Can €36 be labelled “extracted listed total”?

Supported response: No. It can be a separate calculated check only.

Look for this misconception: A plausible calculation counts as document evidence.

Schema: analogy, mechanism and check

Familiar picture: A shipping form checks boxes and value types before accepting a parcel.

Familiar part	Technical meaning
Boxes	field names
Type rule	schema
Reject stamp	validator flag

Where this analogy breaks: A correct shape does not prove the value came from the document.

Actual idea: Validate names and types of returned fields with Python.

Work it aloud: An ID of INV-04 and a listed total of €36 can pass type checks but fail a source check.

Ask before revealing: Which test detects a wrong but well-typed amount?

Supported response: Compare the returned amount with the original invoice text.

Look for this misconception: Valid JSON means the facts are valid.

Abstention: analogy, mechanism and check

Familiar picture: A careful witness says “I did not see that number.”

Familiar part	Technical meaning
Unseen figure	missing field
Honest reply	null
Follow-up	human review

Where this analogy breaks: Abstention can be too cautious; measure missing and falsely missing fields.

Actual idea: A missing value should stay null or cause a review flag, rather than be guessed.

Work it aloud: When the total is absent, the validator flags null for review without inventing a figure.

Ask before revealing: Why preserve a missing value?

Supported response: It separates unknown source content from an independent calculation.

Look for this misconception: Every missing field is a model failure to be hidden.

New situation for discussion

A syntactically valid JSON reply lists total €36. Which two checks disagree?

Memory cue: Specify the task, keep missing fields missing, then check structure and source.

Guided laboratory

Run extraction validation on ten synthetic records.

1. Compare a vague request with a field-specific request
2. Inspect ten synthetic invoice records
3. Validate an extraction dictionary against the schema
4. Explain a failure caused by an invented total

Individual change: Delete one total and confirm the missing-field review flag fires. Expected evidence: Prompt revision and before/after field-check table.

Checks and common error

Measure: Fields correct / fields checked; report missing fields separately. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: Valid JSON means the invoice facts are true.

Correction: Syntax and data types are only the first check; compare values with the original document.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

OpenAI, Evaluation best practices. <https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Google, Develop a generative AI application.

<https://docs.cloud.google.com/docs/ai-ml/generative-ai/develop-generative-ai-application>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Fine-tune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO4, LO5 | 2 hours concepts + 2 hours guided practice

The idea in plain language

A clear extraction prompt specifies the fields, the source and the missing-value rule. Validate the result in layers: syntax, schema, factual support, then business rules. A schema-valid dictionary can still contain a wrong amount.

A worked example

An invoice shows quantity 3 and unit price 12 but no printed total. The extracted listed_total should remain None. A separate calculated_total may be 36. Keeping these fields separate prevents a calculation being misrepresented as transcription.

Practice before reading the answers

1. Write a missing-value instruction for invoice extraction. (4 suggested practice marks)
2. A JSON response contains a valid number for an absent total. Does it pass? (4 suggested practice marks)
3. Why store listed_total and calculated_total separately? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. Use only the provided invoice. Return null for an absent field and do not infer or invent the printed total.
2. It may pass schema validation but fails evidence validation.
3. They answer different questions: what the document says and what arithmetic implies.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

O1 OpenAI: Evaluation best practices

<https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Study focus: Task-specific testing, repeated outputs and separate evaluation cases.

O3 OpenAI Academy: Prompting

<https://academy.openai.com/en/public/clubs/work-users-ynjqu/resources/prompting>

Study focus: Task, context, expected format and iterative improvement.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.